

EXERCISES

Teachers: Catarina Neves

(cneves@novaims.unl.pt)

Daniel Caridade

(dcaridade@novaims.unl.pt)

3. DATA MODIFICATION

Associate a library named *mydata* with the */home/u63609023/P4B/Data* folder (suppose that this is the folder where you keep the files of the exercises).

EXERCISE 1

Consider the *mydata.msales* data set, which contains the *supermarket_sales.txt* data as described in Exercise 1.

- a) Create a new data set that does not contain the variables *City*, *Time* and *Payment*.
- b) Change the data set created in the previous paragraph so that it only contains selected transactions for product lines categorized as 'Health and beauty' or 'Home and lifestyle' with a total price including tax exceeding 200.
- c) In a single Data Step, create a new data set based on *mydata.msales* such that:
 - It has a new variable named *Total_Without_Tax* that corresponds to the total price of each transaction without tax;
 - It only contains the variables: *Invoice_ID*, *Product_Line*, *Tax_5*, *Total*, *Date* and *Total_Without_Tax*;
 - It contains selected transactions for product lines categorized as 'Food and beverages';
 - The *Total* and *Total_Without_Tax* variables are associated with the labels *Total price including tax* and *Total price excluding tax*, respectively.

EXERCISE 2

Consider the *mydata.employees* data set, which has the following structure:

Variable	Description	Type
ID	Identification	Numeric
Surname	Last name	Character
Name	Name	Character
City	City	Character
Sex	Gender	'F' or 'M'
Cod	Department code	Character
Salary	Salary	Numeric
Birth	Date of birth	Date

- a) Create a new data set that includes the male employees who were born after 1 January, 1950.
- b) Considering the *mydata.employees* data set, create a new data set such that:
 - It includes employees with last name starting with A, B or C;
 - It includes a variable that extracts the month from the date of birth of employees.
- c) Suppose employees of departments ME1, ME2 and ME3 were increased in 1100, 1200 and 1300, respectively; and the other departments were increased 1000. Create a new variable, *SalaryA*, which corresponds to this salary adjustment.
- d) Solve the previous paragraph again using an equivalent Data Step.

EXERCISE 3

Import *Churn* dataset that contains data regarding 10,000 clients of a bank.

- a) Create a new data set containing the following new variables:
 - a. *Begin_age* that corresponds to the age of the client when it opened an account in the bank;
 - b. *Salary_while_cust* that corresponds to the estimated salary that the client received since it is a client in that bank (consider that the salary of one year is the 14 times the *EstimatedSalary*);
 - c. *Invest* that represents the percentage of the client's account *Balance* relative to their *Salary_while_cust*.

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- b)** Change the data set created in the preceding paragraph such that:
- It includes a new variable named *rBalance* that corresponds to the rounding of *Balance*;
 - Change the *Geography* variable such that the names appear in upper case;
 - It only contains the variables: *Customerid*, *Geography*, *CreditScore* and *rBalance*;
- c)** We want to divide the clients by alphabetic order. So, create a new data set that only contains clients whose surname starts with A.